YILDIZ TECHNICAL UNIVERSITY

DEPARTMENT OF MATHEMATICS

MATHEMATICAL ENGINEERING
SQL PROJECT

TURKEY BIRD ATLAS

Milestone Technical Report

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1. Problem Statement

Right now, there is no easy way to find information about bird species in Turkey in one place. People who watch birds or study them keep their notes separately. Some use simple spreadsheets. Others write things down in notebooks or share data in online forums. Because of this, it is hard to search for a bird by name, check which region it lives in, or find out if it is endangered.

The goal of this project is to build a SQL database called the Turkey Bird Atlas. This database will store important information about birds found in Turkey. For each species, it will keep the Turkish and Latin names, the bird family, where it lives, whether it migrates, and its protection status. With this database, users will be able to search and filter bird data in a clear and organized way.

2. Why This Project Matters

Turkey is a very important country for birds. It is located between Europe, Asia, and Africa, so many birds fly through Turkey when they migrate each year. More than 480 different species have been recorded here. A good database for these birds would help many different people:

Who benefits	How they benefit
Researchers and birdwatchers	They can search and filter bird records quickly instead of looking through many separate files.
Conservation groups	They can see which species are endangered and track changes in their status over time.
Students learning SQL	Bird data is a good real example to practice joins, views, group by, and other SQL features.
Local authorities	They can use regional data to make better decisions about parks and nature reserves.

From a database point of view, this topic also has interesting relationships to model. A bird can live in more than one habitat. The same bird can appear in many regions. These many-to-many connections made me think more carefully about the table design and how to connect everything properly.

3. Preliminary Schema Design

The database has six tables. I tried to keep each table focused on one thing. To avoid storing the same data more than once, I used bridge tables to handle many-to-many relationships between species, habitats, and regions.

3.1 Core Tables

Table	Key Columns	Type	Purpose
species	species_id PK, name_tr, name_latin UNIQUE, family, order_name, migration_status, conservation_status	Main entity	Holds the main info for each bird species.
habitats	habitat_id PK, name, description	Reference	Lists habitat types: wetland, forest, steppe, coast, mountain.
regions	region_id PK, city_name, nuts_code, geographic_zone, area_km2	Reference	Turkish provinces and their geographic zone.
species_habitat	species_id FK, habitat_id FK, is_breeding (composite PK)	Bridge	Links species to their habitats (many-to-many).
species_region	species_id FK, region_id FK, presence_type, season (composite PK)	Bridge	Shows which species are found in which regions.
observations	obs_id PK, species_id FK, region_id FK, obs_date, observer, latitude, longitude, count, photo_taken	Records	Stores field sightings with date and GPS location.

3.2 Relationships

- One species can be linked to many habitats. One habitat can have many species.
- One species can appear in many regions. One region can have many species.
- One species can have many observation records. Each observation belongs to one species.
- One region can have many observations recorded in it.
- The conservation_status column uses IUCN codes: LC, NT, VU, EN, CR, EX, DD.
- The migration_status column shows if a bird lives here all year or only visits in a certain season.

3.3 Future Changes

For the final version, I want to add a few more tables. A Photos table could store images linked to observation records. A Threats table could list the main dangers for each species. A Sources table would keep track of where the data came from. These changes will make the database more useful.

4. Proposed SQL Features

I plan to use more than just basic SELECT queries. The table below shows which SQL features I want to use and why each one makes sense for this project:

SQL Feature	How I will use it in this project
Primary & Foreign Keys	Every table has a primary key. Foreign keys connect the tables and keep the data consistent.
CHECK / ENUM Constraints	The conservation status field only allows IUCN codes. The migration status is also limited to a set list.
JOINs (INNER, LEFT)	I will join species, habitat, region, and observation tables to get full results in one query.
Views	I will create a view for species summary info and another view to support a map-style display.
Aggregation (COUNT, SUM, AVG)	I can count how many species are in each region or find the total number of birds seen in a period.
GROUP BY + HAVING	I can group data by region and filter to show only regions with a certain number of records.
Subqueries / CTEs	A CTE will help me build a species profile query that is easier to read and follow.
Indexes	I will add indexes on columns used often in WHERE conditions to make queries run faster.
Filtering & Sorting	Users can filter by status — for example, show only endangered birds — and sort by date or count.

5. Example Queries

Query 1 — Show all species with their habitats and regions

```
SELECT s.name_tr, s.name_latin, s.conservation_status,
       GROUP_CONCAT(DISTINCT h.name) AS habitats,
       GROUP_CONCAT(DISTINCT r.city_name) AS regions
FROM species s
LEFT JOIN species_habitat sh ON s.species_id = sh.species_id
LEFT JOIN habitats h         ON sh.habitat_id = h.habitat_id
LEFT JOIN species_region sr  ON s.species_id = sr.species_id
LEFT JOIN regions r          ON sr.region_id = r.region_id
GROUP BY s.species_id;
```

Query 2 — List only threatened species

```
SELECT name_tr, name_latin, conservation_status
FROM species
WHERE conservation_status IN ('VU', 'EN', 'CR')
ORDER BY conservation_status, name_tr;
```

Query 3 — Count observations and photo rate per region

```
SELECT r.city_name,
       COUNT(*)                            AS total_observations,
       SUM(o.count)                        AS total_individuals,
       ROUND(AVG(o.photo_taken) * 100, 1)  AS photo_rate_pct
FROM observations o
```

```
JOIN regions r ON o.region_id = r.region_id
GROUP BY r.region_id, r.city_name
ORDER BY total_observations DESC;
```

Query 4 — CTE: get a summary for each species

```
WITH latest_obs AS (
  SELECT species_id, MAX(obs_date) AS last_seen, SUM(count) AS total_seen
  FROM observations
  GROUP BY species_id
)
SELECT s.name_tr, s.migration_status, s.conservation_status,
       lo.last_seen, lo.total_seen
FROM species s
LEFT JOIN latest_obs lo ON s.species_id = lo.species_id
ORDER BY lo.last_seen DESC;
```

Query 5 — Migratory species found in wetland areas

```
SELECT s.name_tr, s.migration_status
FROM species s
JOIN species_habitat sh ON s.species_id = sh.species_id
JOIN habitats h         ON sh.habitat_id = h.habitat_id
WHERE h.name = 'Sulak alan'
  AND s.migration_status != 'Yerli'
ORDER BY s.name_tr;
```

6. Feasibility and Scope

I think this project is a good fit for the time we have this semester. The topic is clear and well defined. The data I need is available from open websites like eBird and the IUCN Red List. The database is not too big and not too simple. It has enough tables and connections to show different SQL skills, but it is still easy to manage on my own.

By the end of the project, I plan to have the following ready:

- A complete database schema with all tables, primary keys, foreign keys, and constraints.
- Sample data with at least 50 bird species, all 81 Turkish provinces, 5 habitat types, and over 100 observation records.
- At least 2 views: one for species summary information and one for observation data.
- At least 10 queries that use JOINs, aggregation, subqueries, and CTEs.
- Notes on where the data came from and how I collected it.
- A GitHub repository with all SQL files and a short README file.

I chose this topic because I find birds interesting and because the data has many different types of connections. Working on this project helped me understand how relational databases work in a real situation, not just in theory.